



TECHNICAL BRIEF

OCIMS/SOMISANA High Resolution Ocean Model of the South African West Coast (SA-West V1.0):

Configuration Details and User Guide

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Objective – Developed by the SOMISANA team at SAEON’s Egagasini Node as part of the OCIMS initiative, this technical brief describes a high-resolution ocean model of South Africa’s West Coast. It provides daily 5-day forecasts alongside a 32-year hindcast (1993–2025). The brief evaluates model performance and outlines access protocols for dataset outputs and processing tools

Despite continuous efforts to increase ocean observations and improve satellite observing capabilities, a significant paucity of data persists in the oceans surrounding southern Africa. Ocean models are essential as they provide information on the vast and dynamic ocean that is both spatially and temporally cohesive, effectively filling the gaps between sparse observations. By complementing observational datasets, models allow for an improved understanding of ocean processes and their impact on marine ecosystems, weather and climate, extreme events, operations at sea and coastal resilience. Historical ('Hindcast') models are crucial for investigating long-term changes and trends, while forecast models deliver information about a future state of the ocean that supports decision making at various scales, from marine operations to long-term management and mitigation measures.

The freely available SOMISANA models are designed to support both research and operational activities.

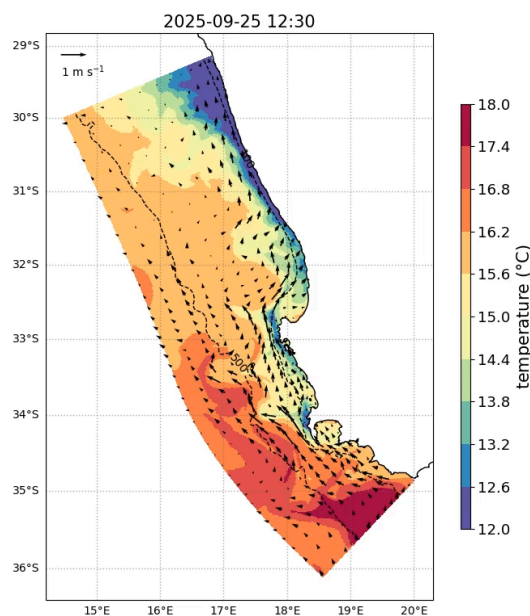


Figure 1: a snapshot of sea surface temperature and current outputs of the SOMISANA SA-West Forecast model for the 25 September 2025.

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List of key acronyms

Acronym	Definition
CROCO	Coastal and Regional Ocean COmmunity Model
CSIR	Council for Scientific and Industrial Research
DeSTs	Decision Support Tools
DFFE	Department of Forestry, Fisheries and the Environment
NRF	National Research Foundation
OCIMS	National Oceans and Coastal Information Management System
SAEON	South African Environmental Observation Network
SAWS	South African Weather Service
SOMISANA	Sustainable Ocean Modelling Initiative: a South African Approach

Introduction

Despite ongoing advancements in ocean observation and satellite monitoring, significant data gaps persist. High-resolution ocean models are essential for bridging these observational gaps as they provide spatially and temporally continuous data that allow us to deepen our understanding of marine environments and how they're changing. Recognizing the need for quality information about the changing state of the ocean to support a sustainable oceans economy, the National Oceans and Coastal Information Management System (OCIMS) provides decision support tools (DeSTs) for effective management and the good governance of South Africa's oceans and coasts. OCIMS is led by the Department of Forestry, Fisheries and the Environment (DFFE) and is implemented with consortium partners: the South African Environmental Observation Network (SAEON), the South African Weather Service (SAWS) and the Council for Scientific and Industrial Research (CSIR). Funded by the OCIMS project to support the consortium objectives, SAEONs SOMISANA (Sustainable Ocean Modelling Initiative: a South African Approach) team develops regionally optimized ocean models that provide information about the changing state of the ocean to support various OCIMS DeSTs.

This technical brief describes the configuration, output quality, value-added products, accessibility and processing tools for a high-resolution ocean model of South Africa's West Coast (SA-West V1.0) that provides both daily 5-day forecasts as well as a 33-year hindcast (1993–2025).

Model description

Overview

All SOMISANA model developments utilize the Coastal and Regional Ocean Community (CROCO) modelling system (<https://www.croco-ocean.org/>) that has been produced specifically to replicate regional, coastal and nearshore ocean processes. It utilizes terrain-following vertical coordinates (sigma-grid) and a free surface, which are crucial for accurately representing shallow coastal shelves and complex bathymetry. CROCO's core strength is its flexibility, including the ability to perform high-resolution simulations using two-way nesting (via AGRIF) and an optional non-hydrostatic solver for very fine-scale phenomena like strong vertical mixing or internal waves. Furthermore, it is designed for multi-model coupling with atmospheric, wave, sediment, and biogeochemistry models, providing the comprehensive physics required for holistic coastal ecosystem and operational forecasting studies. As a regional ocean modelling platform, CROCO models, both hindcast and forecast simulations, follow a downscaling approach, requiring the outputs from a global model to drive its lateral oceanic boundaries.

Grid and bathymetry

The hindcast and forecast versions of the SOMISANA SA-West configuration V1.0 share the same grid. The model curvilinear grid has a variable horizontal grid resolution (1 – 2.5 km) which increases toward the coast (Figure 2). The model has 30 terrain-following (sigma) vertical levels that are built upon a model bathymetry which is interpolated from high resolution digital navigation charts provided by the South African National Hydrographic Office.

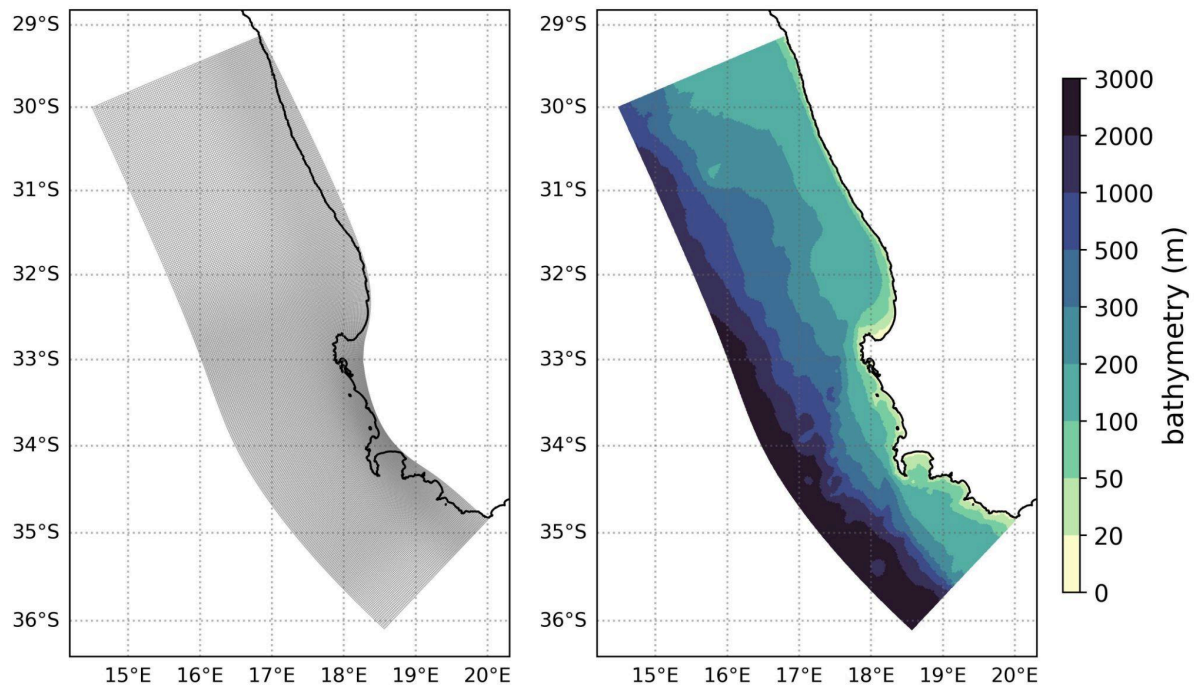


Figure 2: The SOMISANA SA-West V1.0 curvilinear grid configuration

The CROCO ocean model utilizes the Arakawa C-grid for its horizontal layout, a type of staggered grid that strategically separates where different physical variables are calculated. Instead of placing all variables at the center of a grid cell, CROCO places scalar quantities like temperature, salinity, and sea surface height at the cell's center, known as the rho-point. Conversely, the horizontal velocity components (u for east-west/zonal flow and v for north-south/meridional flow) are positioned at the respective faces of the cell, directly perpendicular to the flow they represent. This specific staggering is essential for the model's accuracy, as it allows for precise, second-order calculations of mass, momentum, and energy fluxes across the cell faces, ensuring key physical laws—like the conservation of energy and potential vorticity—are maintained throughout the simulation of regional and coastal ocean dynamics. This staggered grid means that three different sets of latitudes and longitudes coordinates are provided in the output files.

Forcing datasets

Both the hindcast and forecast configurations require datasets from external data providers to force the model, as detailed in Table 1.

Table 1: Summary of datasets used to force the SOMISANA SA-West V1.0 hindcast and forecast simulations.

	Hindcast	Forecast
Surface forcing (momentum fluxes, i.e. the wind field)	ERA5 ¹ , 1993 - 2015 WASA3 ² , 2009-2013	GFS ³ , SAWS ⁴
All other surface forcing parameters	ERA5 ¹	GFS ³ , SAWS ⁴

Open boundary conditions	GLORYS12V1 ⁵	MERCATOR ⁶ , HYCOM ⁷
Tides	Not implemented	TPXO10 ⁸

1. ~ 30 km resolution reanalysis product from ECMWF (Hersbach et al., 2020; Copernicus Climate Change Service, Climate Data Store, 2023;)
2. ~ 3 km resolution WRF Wind Atlas for South Africa dataset provided by the Climate Systems Analysis Group (CSAG) (Hahmann et al., 2021). The outputs of a 5 year simulation are not available via the SOMISANA catalogue, but can be made available upon request.
3. ~ 30 km resolution Global Forecast System from NOAA (<https://www.nco.ncep.noaa.gov/pmb/products/gfs/>)
4. 4.4 km resolution atmospheric forecast model run operationally by the South African Weather Service (SAWS)
5. ~9 km resolution global ocean reanalysis product produced by Mercator Ocean International (Lellouche et al., 2019; <https://doi.org/10.48670/moi-00021>)
6. ~9 km resolution global ocean analysis and forecast product produced by Mercator Ocean International (<https://doi.org/10.48670/moi-00016>)
7. ~9 km resolution global ocean analysis and forecast product produced by the HYCOM consortium (<https://www.hycom.org/dataserver/gofs-3pt1/analysis>)
8. ~4 km resolution global tidal constituents (<https://www.tpxo.net/global/tpxo10-atlas>)

Table 1 indicates that the forecast configuration employs two different open boundary forcing products as well as two different surface forcing products. This allows us to provide a four-member ensemble of forecasts: MERCATOR-GFS, MERCATOR-SAWS, HYCOM-GFS and the HYCOM-SAWS which serves two purposes - greater spread of potential forecast ocean states and increased redundancy in the operational system.

Parameterisations

The hindcast and forecast versions of the SOMISANA SA-West configuration V1.0 share the same model parameterizations, as detailed in Table 2.

Table 2: Parameterisation details for the SOMISANA SA-West configuration V1.0 (capitalised keys in brackets refer to the CPP key names used in compiling the CROCO source code for this configuration)

CROCO version	1.3.1
Horizontal grid	Curvilinear (128 x 446 grid points)
Vertical grid	Sigma (30 vertical levels)
Horizontal advection: momentum	3rd-order upstream-biased (UV_HADV_UP3)
Horizontal advection: tracers	Rotated-Split 3rd-order upstream (TS_HADV_RSUP3)
Vertical advection: momentum	Spline vertical advection (UV_VADV_SPLINES)
Vertical advection: tracers	Spline vertical advection (TS_VADV_SPLINES) + adaptive semi-implicit vertical advection (VADV_ADAPT_IMP)
Horizontal mixing	Disabled - no explicit lateral viscosity or tracer diffusion
Vertical mixing	Generic Length Scale closure, k-epsilon (GLS_MIXING)

Barotropic open boundary conditions	Characteristic scheme for 2D momentum (OBC_M2CHARACT)
Baroclinic open boundary conditions	Orlanski radiation for 3D momentum (OBC_M3ORLANSKI) and tracers (OBC_TORLANSKI)
Air-sea fluxes	COARE 3.0 bulk formulae (BULK_FLUX)
Tides	Disabled in the hindcast, enabled in the forecast (TIDES)
Rivers	Disabled

Model evaluation

Quality testing is conducted during both the hindcast development phase as well as on the forecast system. While the considerations and objectives for each phase of evaluation differ slightly, both provide a quantitative assessment of the skill of the model.

The development of the operational forecast system (OFS) begins with a model configuration that is run in hindcast mode that allows for thorough assessment of how well the model configuration and parameterizations capture the general dynamics as well as various scales of variability, trends and episodic events during a period in the past. During this phase model ‘tuning’ is carried out in order to optimize the configuration and potentially also test different forcings. In the case of the SA-West V1.0 configuration, two different versions of the simulation were evaluated: one that was forced using the ERA5 surface momentum fluxes (winds) and the other with the higher resolution WASA3 momentum fluxes. This procedure tests how surface forcing resolution affects model skill. Comparing archived forecasts with satellite and in situ observations we are able to test the forecast skill of the model. This allows us to assess how the skill degrades as the forecast window gets larger.

Our model evaluation protocol includes assessing our model outputs (both hindcast and forecast) with respect to a state of the art global system that provides historical and forecast information (GLORYS, that we also use for lateral boundary conditions). Our aim is to produce regionally optimized models that perform better than the freely available global systems

Our model evaluation process aligns with recognized standards and best practices that are described in Alvarez Fanjul et al., 2022 and Alvarez Fanjul 2024. While our evaluation procedure involves all metrics, as defined in Hernandez et al., 2009, in this summary of our evaluation results we focus primarily on Class 4 metrics that relate to the calculation of various skills score metrics of model-observation comparisons including model bias, root mean square error/difference and correlation (described below).

$$Bias = m - o$$

$$RMSE = RMSD = \sqrt{\frac{\sum_{i=1}^n (m_i - o_i)^2}{n}}$$

$$r = \frac{\sum (m_i - \bar{m})(o_i - \bar{o})}{\sqrt{\sum (m_i - \bar{m})^2} \sqrt{\sum (o_i - \bar{o})^2}}$$

Where m and $\bar{m}$ are model results and long term means respectively and, similarly, o and $\bar{o}$ are observations and observational long term mean. r is the Pearson correlation coefficient (+1 and -1: perfect correlation and anti-correlation respectively).

In this user-guide we show only a selection of the evaluation assessments. For all evaluations (including current and temperature profiles) refer to the full evaluation report that can be accessed through the SOMISANA catalogue (<https://catalog.somisana.ac.za/>).

Datasets used for validation

Various in situ and satellite observational datasets were utilized for the SA West hindcast validation and forecast skill assessment. All of the datasets used for the complete validation are described below, but only a selection of the results are shown in this report.

In situ Observations

Acoustic Tracking Array Platform (ATAP)

This dataset was provided by the NRFs South African Institute for Aquatic Biodiversity (SAIAB), an official partner of the Canadian-based global Ocean Tracking Network (OTN). They provide hourly temperature measurements at a number of acoustic stations around the coast (<https://saiab.ac.za/platforms/acoustic-tracking-array-platform/>) at depths that vary from 4 to 55 m.

Coastal Underwater Temperature Recorders (UTRs)

Long-term coastal UTR observations from various sources, collated for 87 sites around the coast, were provided by AJ Smit (UWC). The dataset is fully described in Smit et al., (2013). All sites within the SA-West model domain are surface, or near-surface, and were used for evaluation purposes.

St. Helena Bay Temperature Moorings

Data from two temperature moorings, deployed at 20 m and 70 m depths off Elands Bay (32.292°S, 18.318°E) and spanning November 2008 to December 2011 and January 2012 to December 2012 respectively (Pitcher et al., 2014).

St Helena Bay Wire-Walker and Acoustic Current Profiler

High frequency temperature and velocity profiles from wirewalker wave-powered profilers and bottom mounted ADCPs for the period February to April 2011. For the location refer to Figure 3. The dataset is described in full in Lucas et al., (2014).

Position of in situ observations used for SA-West model evaluation

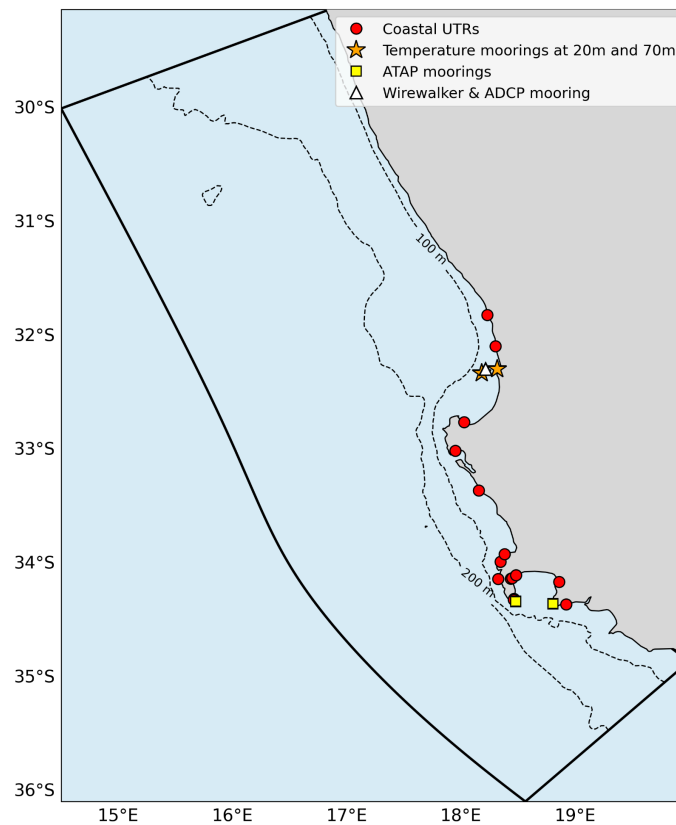


Figure 3: In situ stations within the SA-West domain used for model validation

Satellite Observations

Satellite observations are used to assess how well the model captures large-scale hydrodynamic features.

Sea Surface Temperature

The Operational Sea Surface Temperature and Ice Analysis (OSTIA; Good et al., 2020 and Donlon et al., 2012) provides daily, gap free maps of sea surface temperature at a resolution of $0.05^\circ \times 0.05^\circ$ horizontal grid resolution for the period 17 January to present. OSTIA uses satellite data provided by GHRST (The Group for High Resolution Sea Surface Temperature) together with in situ observations. It can be accessed via Copernicus Marine Services (<https://doi.org/10.48670/moi-00165>).

Sea Surface Height

The DUACS (Data Unification and Altimeter Combination System) processing system provides daily, gap-free Level-4 (L4) maps of Sea Surface Height (SSH) and derived variables at a global horizontal resolution of $0.125^\circ \times 0.125^\circ$ for the period 4 October 2022 to present. DUACS merges multi-mission satellite altimetry data with external auxiliary datasets and orbit corrections. It can be accessed via Copernicus Marine Services (<https://doi.org/10.48670/moi-00149>).

Hindcast evaluation

Comparison against satellite observations

In order to assess the skill of the model in capturing the spatial patterns as well as the variability of sea surface temperature, we produce the standard Class 4 metrics (Bias, RMSE and Correlation: Figures 4, 5 and 6 respectively)) for surface maps of our regionally optimized SA-West domain, as well as for GLORYS (global product used for forcing the lateral boundaries of our model). Due to the intense seasonal signal in the Benguela system, these metrics are computed for each season.

The GLORYS model displays a warm bias throughout the year ($>2^{\circ}\text{C}$ in places) that is most intense during summer/spring months (DJF, MAM), the upwelling season, between Cape Point ($\sim 34^{\circ}\text{S}$) and Cape Columbine (including the region of the Columbine upwelling plume) at about 33°S . A narrow strip of cool bias is also a feature of the GLORYS summer SSTs, but this is limited to the northeast corner of False Bay, $\sim 31.5^{\circ}\text{S}$ and $\sim 29^{\circ}\text{S}$. Both the ERA5 and WASA3 forced versions of the SA-West configuration have smaller biases than GLORYS, during both summer and winter months. In addition, they have similar patterns of SST bias: warm offshore biases during summer months and cool nearshore biases, and during winter months lower biases in general, that tend to be negative (cool) other than in the far north of the domain where there is a slightly warm bias. Despite these similar patterns, the SA-West configuration forced with the WASA3 wind field tends to have smaller biases.

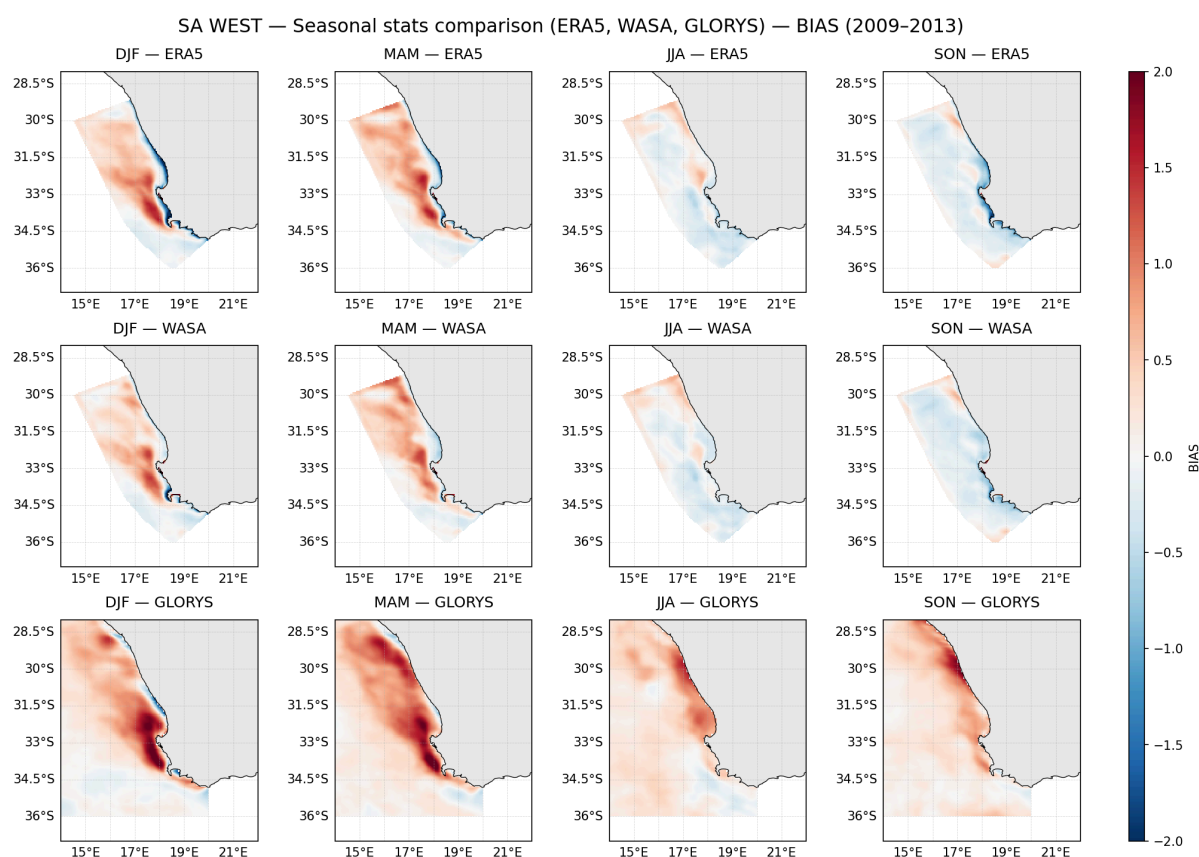


Figure 4: Seasonal SA-West (ERA5: top), SA-West (WASA3: middle) and GLORYS (bottom) model SST biases based on OSTIA satellite SST observations for the period 2009-2013.

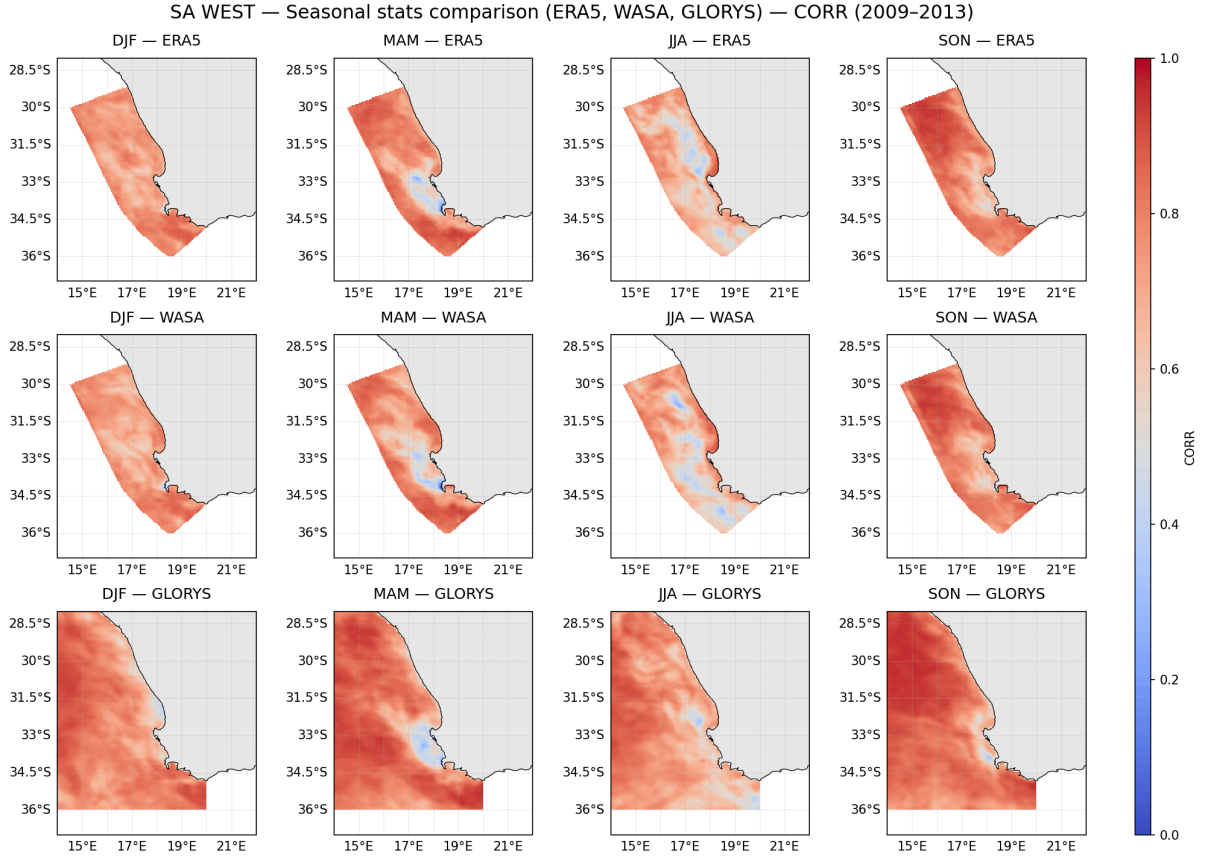


Figure 5: Seasonal SA-West (ERA5: top), SA-West (WASA3: middle) and GLORYS (bottom) model SST correlations with OSTIA satellite SST observations for the period 2009–2013.

The correlation maps of each model dataset relative to satellite SST (Figure 5) reveal how well the models predict fluctuations, but does not reveal anything about the amplitude of the error. On the other hand, the root mean square error (RMSE: Figure 6) tells us how well the models capture the amplitude of the fluctuations, but does not tell us about whether the fluctuations are in phase or not. In this case, we note that, at the large scale, the global model (GLORYS) performs better with respect to correlations (i.e. with values closer to 1) than the SA-West configurations, particularly further away from the coast and in intensely frontal regions. The cRMSE in all models (GLORYS and the SA-West configurations) is higher in regions of intense frontal activity (e.g the Benguela Jet, the Cape Columbine Plume, the coastal upwelling jet etc..) and lower in regions of less turbulence. The SA-West models have slightly lower cRMSE in the frontal regions than GLORYS, which suggests that though they do not capture the timing of the SST fluctuations as well as GLORYS (having lower correlations), they capture the amplitude of the fluctuation slightly better.

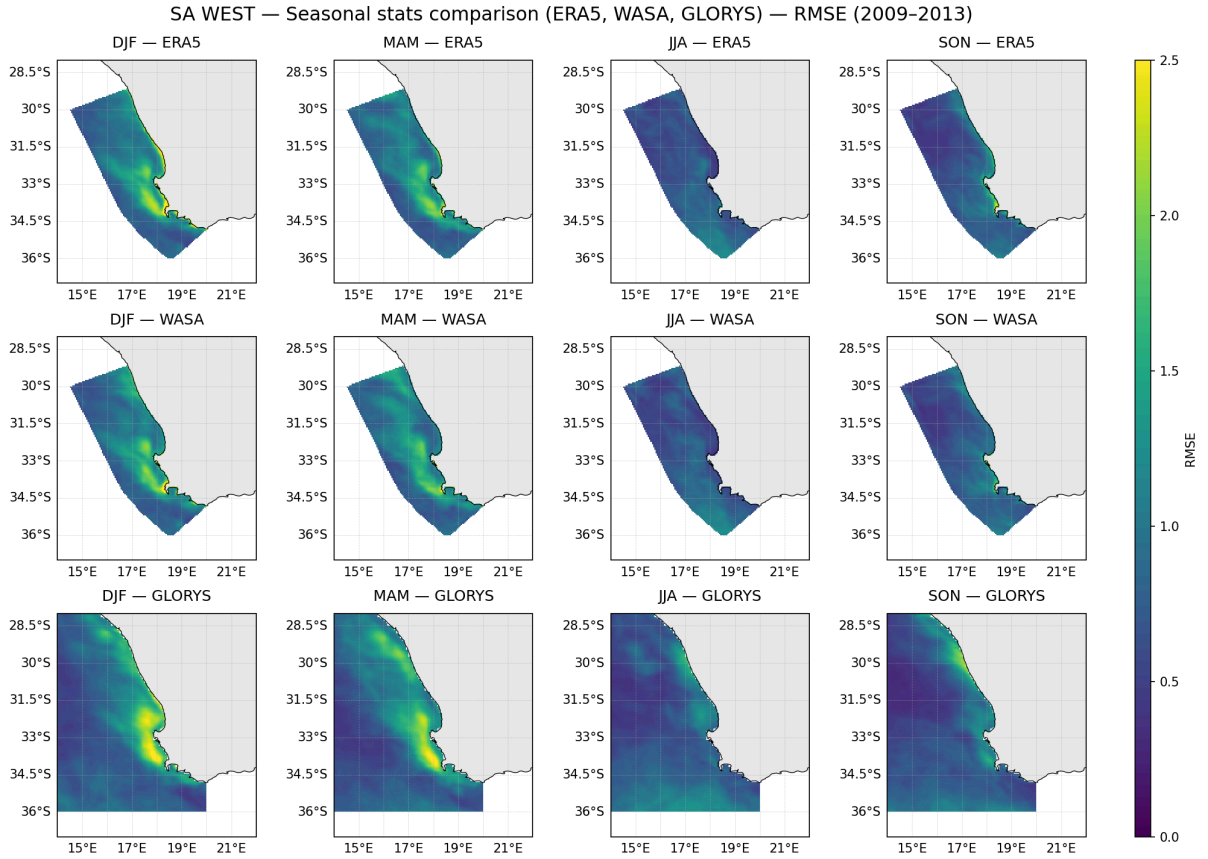


Figure 6: Seasonal SA-West (ERA5: top), SA-West (WASA3: middle) and GLORYS (bottom) model SST RMSEs relative to OSTIA satellite SST observations for the period 2009–2013.

Comparison with in situ observations

The evaluation plots chosen for this summary include an assessment of the coastal temperatures within the Cape Point upwelling cell (Kommetjie) and a station within St Helena Bay (Lamberts Bay). The location of these stations can be found in Figure 3.

The observed, climatological (seasonal mean) coastal temperatures off Kommetjie (blue line in Figure 7c) display a distinct seasonal cycle, with coldest temperatures during the summer upwelling season. Both of the SA-West configurations (red and green line: ERA5- and WASA-forced respectively) capture this seasonal cycle, albeit with a warm bias of, on average, $\sim 1.25^{\circ}\text{C}$ during summer months (November to April) and $\sim 0.25^{\circ}\text{C}$ during winter. The GLORYS model shows a persistent warm bias that is highest during summer months ($\sim 2.75^{\circ}\text{C}$) and decreases ($\sim 1.5^{\circ}\text{C}$) during winter months and it is unable to capture the seasonal cycle: cooler coastal temperatures during summer than winter. The correlations and cRMSEs are calculated for each model, relative to the in situ observations for the period 2009–2014, for the total temperature as well as the anomalies (temperature - climatological mean) (Figure 7a and b respectively). Both SA-West configurations have higher correlations and lower cRMSEs than GLORYS for both of the timeseries, with the SA-West forced with the higher resolution wind field (WASA3) only having a very slightly higher and lower correlation and cRMSE respectively (i.e. performing only marginally better than the ERA5 forced configuration).

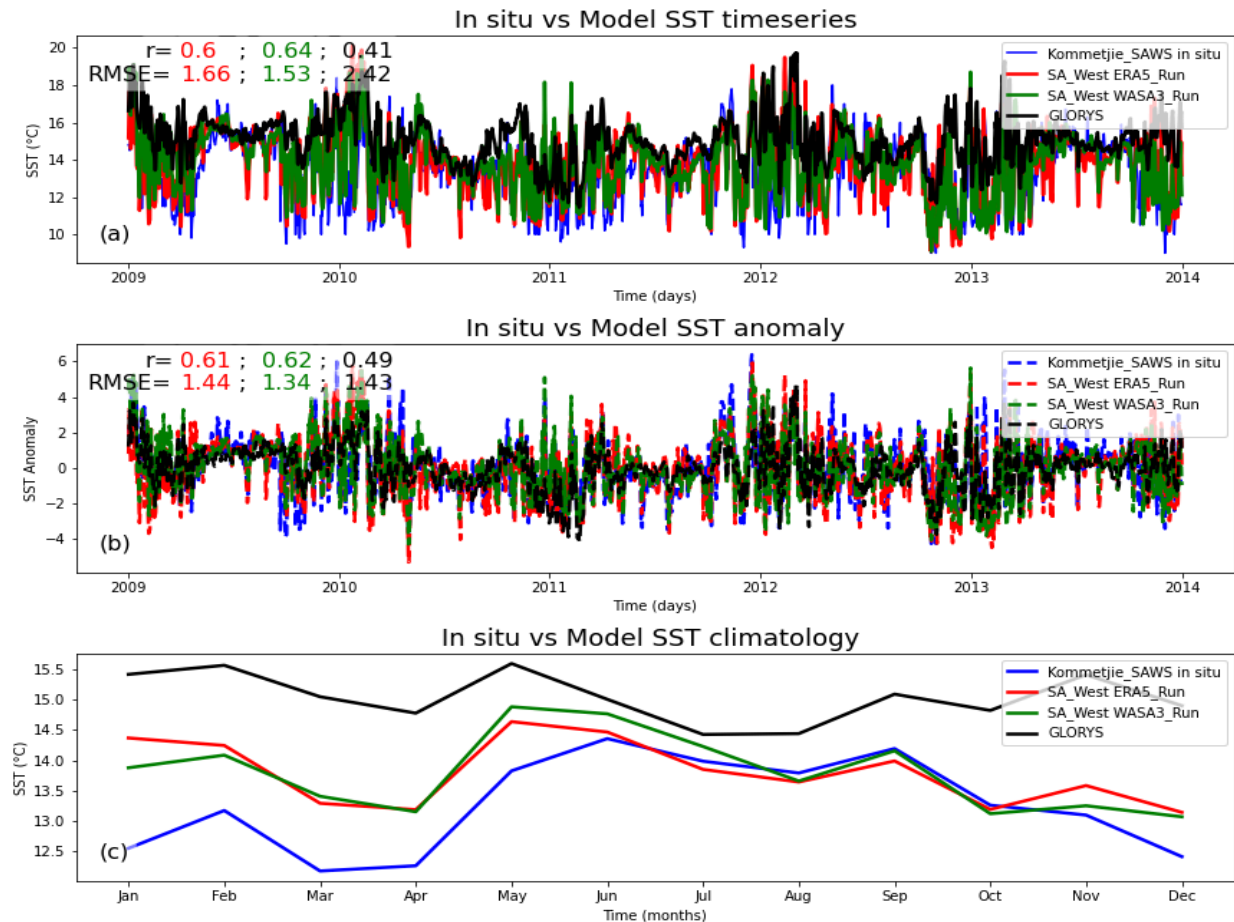


Figure 7: Kommetjie coastal temperatures (a), anomalies (b) and climatology (c) spanning the period 2009-2014, from in situ observations (blue) and based on the SA-West ERA5 (red), SA-West WASA3 (green) and GLORYS (black) models.

In Lamberts Bay (Figure 8) the seasonal cycle of temperature based on in situ observations reflects slightly warmer summer temperatures than during winter. While all of the models (GLORYS as well as the SA-West configurations) reflect this, they all have biases, with GLORYSs being the most persistent of, on average about $+1.5^{\circ}\text{C}$ throughout the year. The SA-West ERA5 configuration performs well between June and October, with insignificant biases but between November and May has a persistent cool bias that is most pronounced during peak summer in January, with a bias of -2°C . The SA-West WASA3 configuration has a minimal warm bias between April and December ($<0.25^{\circ}\text{C}$ on average), that increases during the peak upwelling season between January and March to about $+1.5^{\circ}\text{C}$. In terms of capturing the variability of nearshore temperatures, the SA-West configurations both have lower RMSEs and higher correlations with in situ observations than the GLORYS model. In this location the SA-WEST WASA3, forced with higher resolution winds, performs best with significantly higher correlations and lower RMSEs.

Figures 7 and 8 demonstrate that a simple regional downscaling to higher horizontal resolutions from a state of the art global product improves the nearshore model solution for temperature. They additionally demonstrate that regions with complex wind structures such as St Helena Bay (e.g the wind shadow in the lee of capes) particularly require a high resolution wind field to improve the model skill.

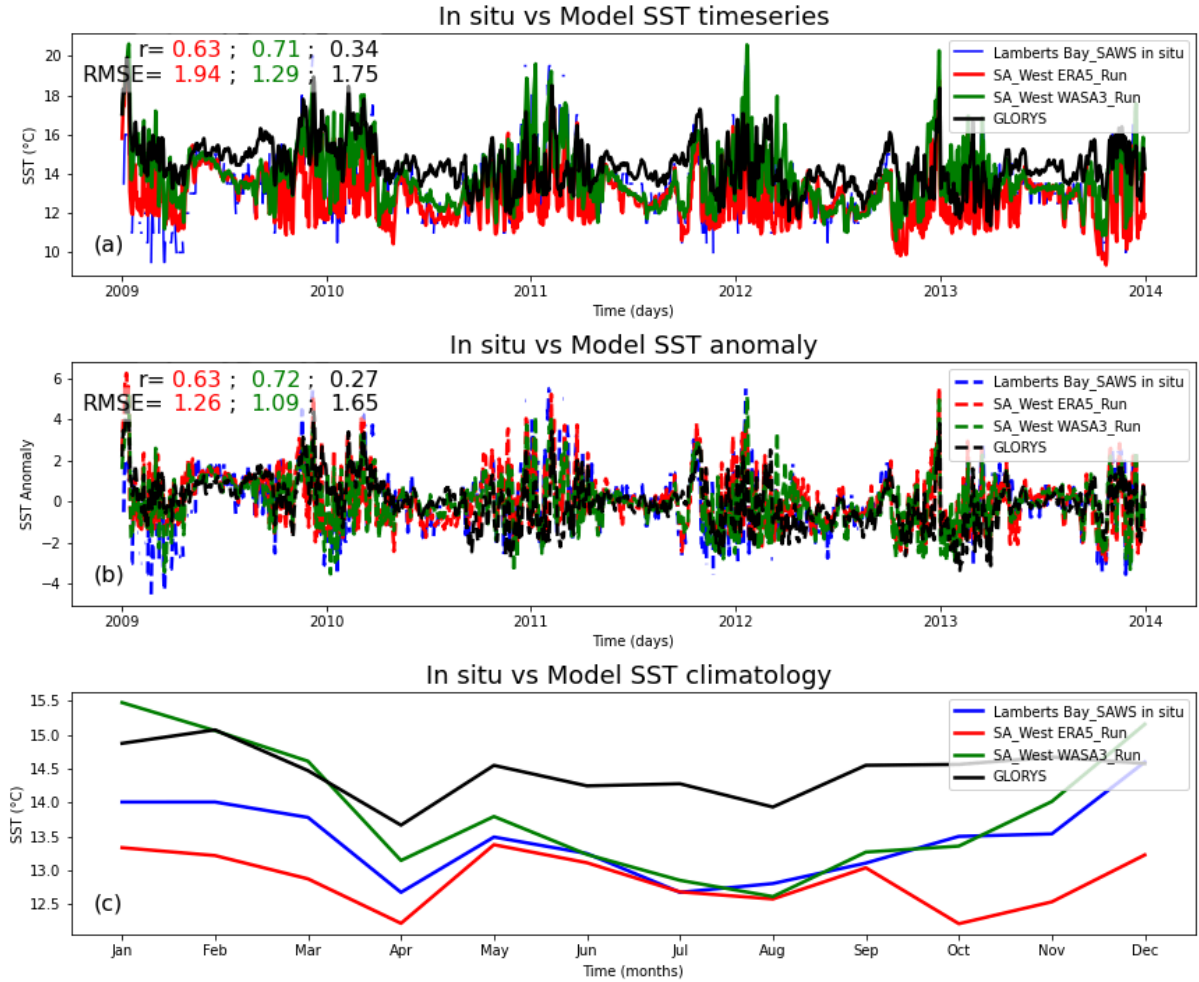


Figure 8: Lambert's Bay coastal temperatures (a), anomalies (b) and climatology (c) spanning the period 2009-2014, from in situ observations (blue) and based on the SA-West ERA5 (red), SA-West WASA3 (green) and GLORYS (black) models.

Forecast skill

As well as evaluating the hindcast simulation of the SA-West configuration, we test how well it captures the forecast. This requires archiving our forecast outputs, or running the forecasts for some historic period so that the skill of the forecast windows can be assessed with respect to satellite or in situ observations for that period. The four different versions (i.e. ensemble members) of the forecast simulation are described in Table 1. In this summary of the forecast skill analysis, we show select results of the following two members: 1) Mercator lateral boundary forcing and GFS atmospheric forcing (SA-West MERCATOR-GFS) that provides a 5-day forecast and 2) Mercator lateral boundary forcing and SAWS atmospheric forcing (SA-West MERCATOR-SAWS) that provides a 3-day forecast.

Comparison against satellite observations

Figure 9 presents instantaneous maps of sea-surface height (red contours) and SSH-derived geostrophic currents (black arrows) from satellite (left column), alongside the model-forecast derived equivalents: MERCATOR-GFS (centre column) and MERCATOR-SAWS (right column), every 2 days between the 27 October and the 31 October.

The Benguela Jet, characterized by an intense SSH front and northwestward flow along the shelf-edge (~500 m isobath) between Cape Point and Cape Columbine, is a key feature of the Southern Benguela with significant ecological importance (Figure 9). In the forecast simulations, the intensity and path of the Benguela Jet as well as the recirculation within St Helena Bay are well captured up to day 3 of both of the forecast runs (centre and right columns, Figure 9). By day 5 of the MERCATOR-GFS forecast, the Benguela Jet becomes more intense than observations and it veers inshore more abruptly just south of Cape Columbine than in observations. This analysis indicates that the large-scale flow is well represented in the forecast models, especially with the shorter forecasts. The accuracy starts to diminish for the longer forecasts.

Comparison against in situ observations

Daily subsurface temperature measurements at Cape Hangklip are compared against the global MERCATOR forecast product, the SA-West MERCATOR-GFS and SA-West MERCATOR-SAWS forecasts (Figures 10 and 11 respectively for the latter two). The analysis is based on time periods that are limited to the duration of overlapping data periods for the forecast simulations and in situ observations: the MERCATOR-GFS comparison from 17 August to 5 October 2024, and the MERCATOR-SAWS comparison from 14 September to 5 October 2024. Figures 10 and 11 compare SA-West forecast members (a) and MERCATOR (b) forecasts (coloured lines: each colour representing a different forecast day) with in situ measurements (black line) recorded at Cape Hangklip (see circle in Figure 9). The bias between the forecast model and the in situ temperature measurements is displayed at the bottom left of each time series.

Taylor diagrams, shown adjacent to the time series, summarise key statistics (standard deviation, correlation coefficient, and centred root mean square error). The black dot in the Taylor diagram represents the reference temperature measurements, while the coloured markers indicate forecast lead times. The closer the coloured marker is to the black line and marker, the greater the skill of the forecast simulation.

Figures 10 and 11 show that the SA-West forecast runs resolve the variability of subsurface temperature more accurately than MERCATOR, as they follow the in situ measurements more closely. This results in a smaller mean bias (bottom left of the time series) and improved performance in the Taylor diagram, where the SA-West markers are generally closer to the reference measurements than those of MERCATOR. Moreover, correlations are higher for the SA-West MERCATOR-GFS forecasts, and the centred root mean square difference (RMSD) values are lower, indicating that the model better captures the trends in the in situ measurements with reduced error compared to MERCATOR. This improvement is less evident in the SA-West MERCATOR-SAWS forecasts, likely due to the smaller number of available runs, resulting in less robust statistics. However, even this short time series shows promising results.

Overall, the comparisons indicate that the SA-West forecasts provide an improved representation of subsurface temperatures relative to MERCATOR and that there is only minimal skill degradation over the 5-day forecast window.

The full forecast skill report can be found via the SOMISANA catalogue:

<https://catalog.somisana.ac.za/catalog/>.

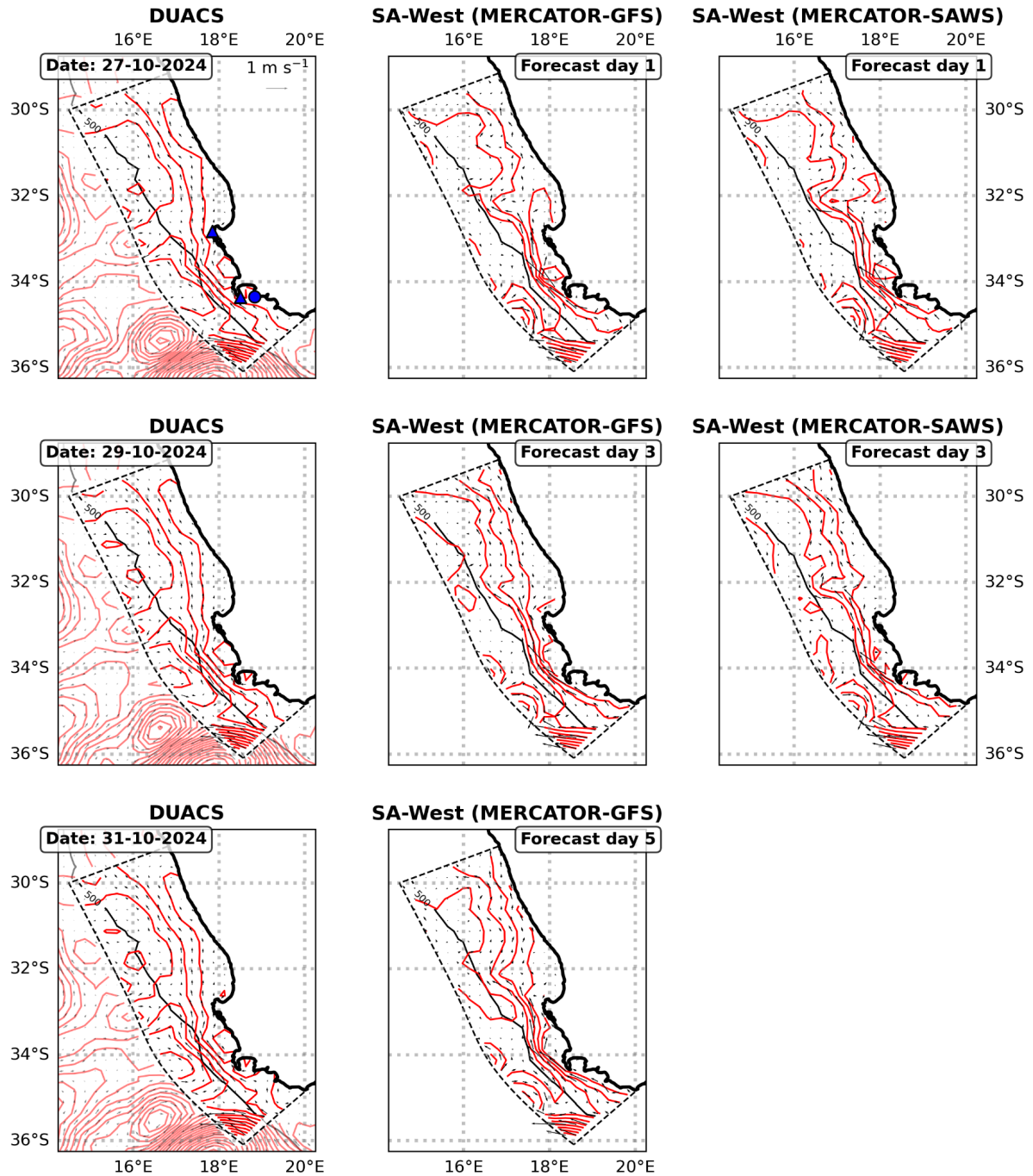


Figure 9: Snapshots of sea-surface height (red contours) and surface derived geostrophic currents (black arrows) comparing DUACS satellite altimetry (left column) to the SOMISANA SA-West Forecast models. The center and right columns are the MERCATOR-GFS and MERCATOR-SAWS forecast runs, respectively. The forecast runs were initialised on 16 August 2024. In the top left figure, triangles show the position of Cape Columbine in the north and Cape Point in the south and the circle the position of the Cape Hangklip UTR.

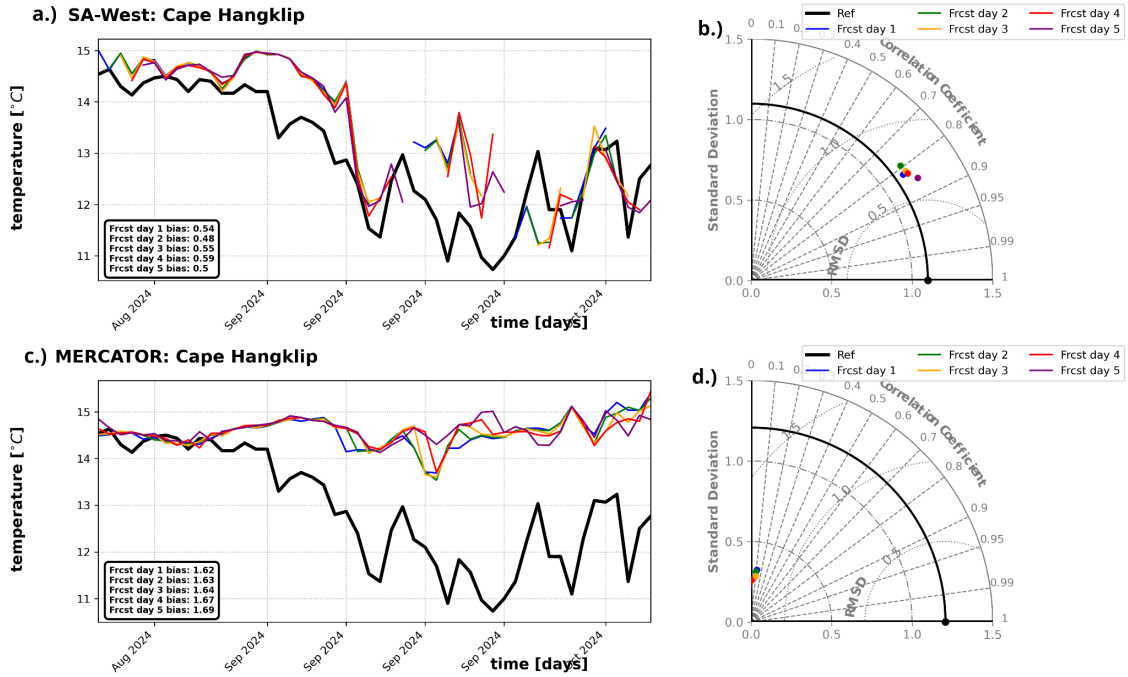


Figure 10: Subsurface temperature time series at the Cape Hangklip UTR from 17 August 2024 to 5 October 2024, showing the SA-West (MERCATOR-GFS, top) and MERCATOR (bottom) forecast models. In situ temperature measurements are shown as the black time series, while coloured time series represent forecast days from the models. Mean biases between the forecast and the in situ temperature measurements are shown on the left corner of the figure. The adjacent Taylor diagram compares model performance (coloured markers) against the reference UTR (black marker) in terms of standard deviation (radial distance from the origin), correlation coefficient (angle from the x-axis), and centred root mean square difference (RMSD; distance between model and reference points).

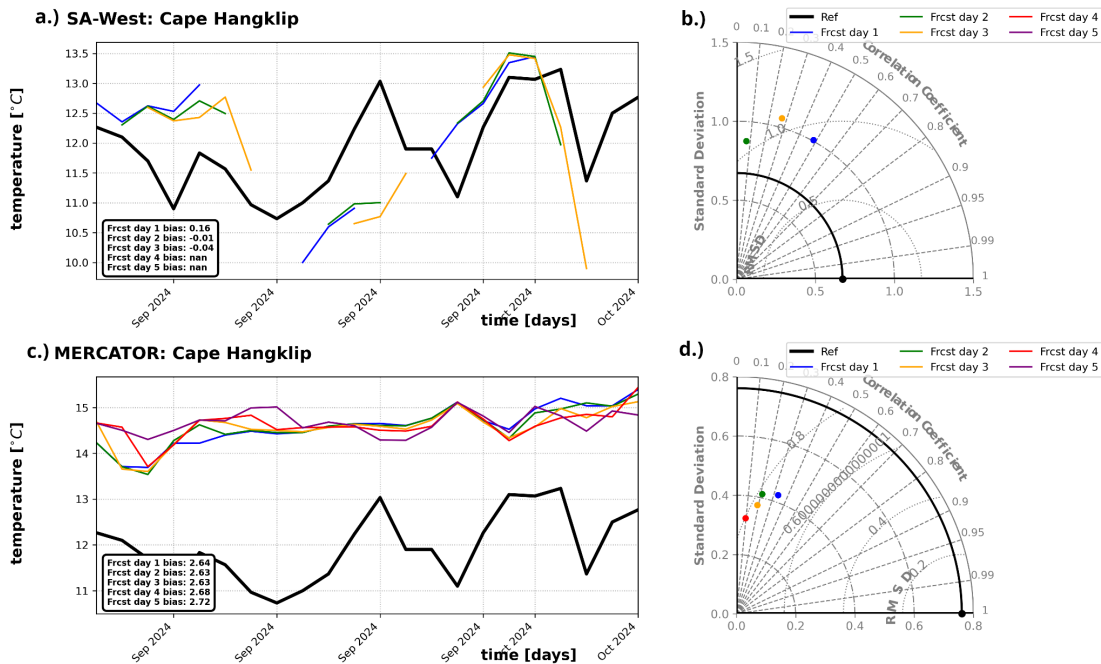


Figure 11: As Figure 10 but at the Cape Hangklip UTR from 14 September 2024 to 5 October 2024, showing the SA-West (MERCATOR-SAWS, top) and MERCATOR (bottom) forecast models.

Model outputs

The available outputs from the hindcast and forecast versions of the SOMISANA SA-West configuration V1.0 are detailed in Table 3.

Table 3: Summary of model outputs for the SOMISANA SA-West V1.0 hindcast and forecast simulations

	Hindcast	Forecast
Run duration	GLORYS-ERA5: 27 years (1993 - 2025) GLORYS-WASA3: 5 years (2009-2013)	All members: 5 days into the future, updated daily
Output frequency	Daily averages for the full water column, hourly averages for the surface layer	Hourly averages for the full water column
Variables	Model-derived: temperature, salinity, current components (u,v,w), sea surface height, vertical diffusivity External forcing: All atmospheric forcing variables	Model-derived: temperature, salinity, current components (u,v,w), sea surface height, vertical diffusivity External forcing: All atmospheric forcing variables (except SAWS momentum fluxes)
Re-gridded outputs ¹	tier 1	tier 1, tier 2, tier 3

1. Refer to section below

Raw CROCO outputs

The raw output from the CROCO simulations are netcdf files, however these files are not cf-compliant and require knowledge of the model architecture in order to be properly used.

Variables are provided on their native Arakawa C-grid, whereby scalar quantities like temperature, salinity, and sea surface height are defined at the cell's center (known as the rho-point), while horizontal vector components (e.g. u, v) are positioned at the respective faces of the cell, directly perpendicular to the flow they represent. This results in three different sets of 2D horizontal coordinates: lon_rho and lat_rho (for scalar quantities); lon_u and lat_u (for u-component of vectors); lon_v and lat_v (for v-component of vectors).

The fact that the grid of this configuration is not regular, but rather a curvilinear grid, means that the horizontal vector components (e.g. u, v, sustr, svstr) represent grid-aligned components rather than zonal/meridional components.

The vertical grid uses terrain-following vertical coordinates (sigma-grid), whereby the depth of each model level is a function of the total water depth, and therefore changes at each time-step. The stretching parameters (theta_s, theta_b, hc) allow for enhanced resolution for the near-surface and near-seabed layers. The depth of each grid cell is however not provided in the output file and must be manually calculated when working with the raw outputs.

The time dimension of the raw output files is not cf-compliant and provided merely as seconds “since initialization”. For the hindcast simulations the initialisation time is 1993-01-01, while for the forecast simulations it is 2000-01-01.

Given the idiosyncrasies of the raw CROCO model outputs, they are not provided directly via the SOMISANA product catalogue, but are archived and can be made available upon request. When working with the raw model outputs, it is recommended that the appropriate postprocessing code be used such as the official croco-tools repositories (<https://www.croco-ocean.org/download-and-documentation/>). Alternatively, the SOMISANA modelling team has developed a python environment which can be used to easily work with the raw CROCO outputs (<https://github.com/SAEON/somisana-croco/wiki/2.-Postprocessing-CROCO-model-output>).

Re-gridded outputs

Three tiers of re-gridded outputs have been developed to provide cf-compliant versions of the model output files in which the idiosyncrasies of the raw CROCO model output files described above have been handled. For reference, the regridding code is contained in the somisana-croco github repository (https://github.com/SAEON/somisana-croco/blob/main/crocotools_py/regridding.py).

Tier 1 outputs

The tier 1 output is provided for both hindcast and forecast simulations. The outputs is provided as cf-compliant versions of the raw CROCO output files, with the following notable amendments:

- The horizontal velocity components (u,v) are interpolated to the scalar (rho) grid so that scalar variables and vector components are on a common grid
- The horizontal velocity components (u,v) have been rotated to be geographically-aligned components (i.e. meridional and zonal) instead of grid-aligned components
- A 'depth' variable is provided which provides the depth below the surface for each grid cell and each sigma level at each time-step

The rotation of the horizontal velocity components is carried out using

$$u_{geo} = u_{grid} * \cos(a) - v_{grid} * \sin(a)$$

$$v_{geo} = v_{grid} * \cos(a) + u_{grid} * \sin(a)$$

where u_{geo} and v_{geo} are geographically aligned vectors (zonal and meridional respectively), u_{grid} and v_{grid} are the grid aligned vectors and a is the angle of grid rotation provided in the output files.

Tier 2 outputs

Tier 2 outputs are identical to Tier 1 outputs with one exception - the outputs are interpolated to specific constant z-levels. This tiered output is only provided for the operational forecast outputs using the following constant depth levels (in meters): 0, 5, 10, 20, 50, 100, 200, 500, 1000.

Tier 3 outputs

Tier 3 outputs are identical to Tier 2 outputs with one exception - the outputs are interpolated to a constant horizontal grid. This tiered output is only provided for the operational forecast outputs using a constant horizontal grid spacing of 0.01 degrees in both longitude and latitude dimensions. The intention of these files is purely for ease of visualisation, given the output is on a rectilinear grid rather than a curvilinear one. These files are not recommended for use in any analysis or research purposes.

Value-Added Products

Hindcast

Climatology

A daily climatology of all model variables throughout the water column is calculated based on the 1993-2025 hindcast SA-West (ERA5) simulation. Following the method defined by Hobday et al. (2016), the climatology was computed for each day using all values that fell within a 11-day window (5 days forward, 5 days back). The climatology was smoothed with a 31-day moving average. The output is formatted as Tier 1 and provided as a netcdf file.

Extremes

Daily 90th and 10th percentile values, based on the 1993-2025 SA-West (ERA5) hindcast, for all variables throughout the water column are formatted as Tier 1 outputs and provided as a netcdf file.

Marine Heat Waves (MHW) and Cool Spells (MCS)

Daily MHW and MCS category flags (see category descriptions below) are provided throughout the water column for the 33 year period of the SA-West ERA5 hindcast simulation (in Tier 1 format). They are calculated based on the 90th and 10th percentile calculations (see above) and identification methods described in Hobday et al., 2016 and Schlegel et al. 2021 respectively: temperatures above the 90th percentile (or below the 10th percentile) that persist for more than 5 days with breaks of no more than 2 days. Positive values correspond to MHW categories, and negative values to MCS categories. The output is provided as Tier 1 formatted netcdf files.

		MHW (+ve flag)	MCS (-ve flag)
Category I	Moderate	1× to 2× standard deviation above 90th percentile	1× to 2× standard deviation below 10th percentile
Category II	Strong	2× to 3× standard deviation above 90th percentile	2× to 3× standard deviation below 10th percentile
Category III	Severe	3× to 4× standard deviation above 90th percentile	3× to 4× standard deviation below 10th percentile

Category IV	Extreme	>4× standard deviation below 10th percentile	>4× standard deviation above 90th percentile
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Stratification

We provide a daily proxy for stratification, for the full 1993 - 2025 period, based on the difference between the potential density at 5 m and 50 m (or bottom in shallower regions). Smaller values represent more mixed conditions, with a value of 0 representing a perfectly mixed column of water. This is also provided as a daily climatology and 90th and 10th percentile values, based on the same period. All are provided as Tier 1 netcdf files.

Forecast

Temperature anomalies

Forecast temperature anomalies are computed operationally based on the daily climatology derived from the SA-West V1.0 hindcast simulation spanning the period 1993-2025. The computation is performed across the full water column and provides a 5-day hindcast and a 3- to 5-day forecast. Figure 12 (left) presents an example of the anomaly product that is routinely generated and made available to end users via downloadable mp4's, the interactive visualisation portal and incorporated into the raw, tier 1, tier 2 and tier 3 product netcdf files.

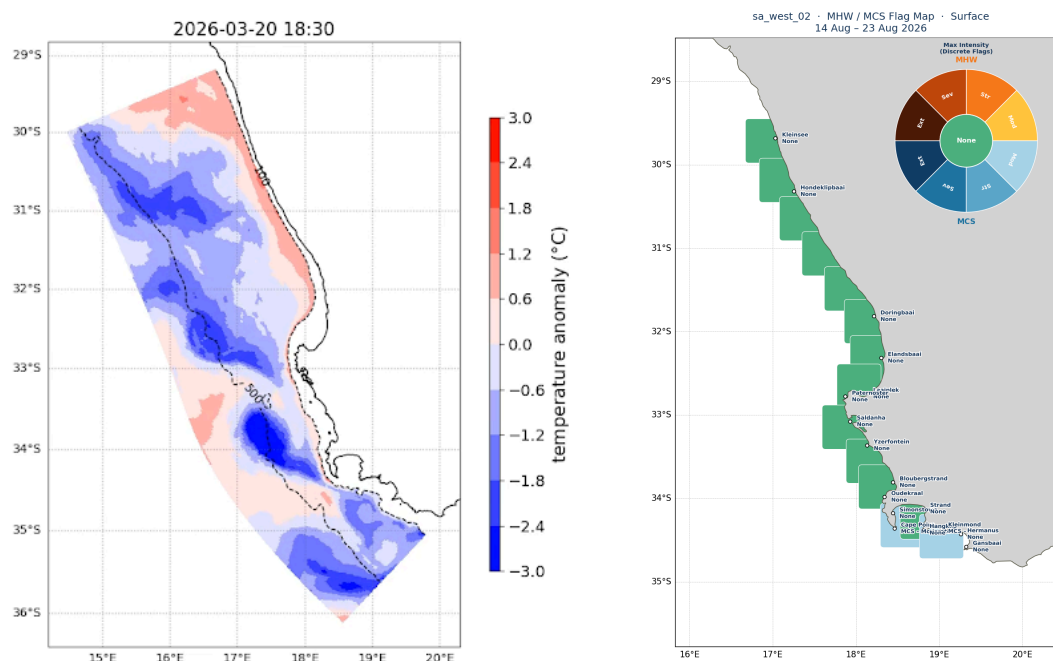


Figure 12: Temperature anomaly on 20 March 2026 (SA-West MERCATOR-GFS) at 100 m (left) and Marine Heat Wave coastal flags produced for the period: 14 - 23 August 2026, based on the SA-West MERCATOR-GFS ensemble member.

Marine Heat Waves (MHW) and Cold Spells (MCS)

Marine Heat Waves (MHW) and Marine Cold Spells (MCS) are computed based on the 1993-2025 SA-West (ERA5) climatology, 90th and 10th percentile values, according to methods described in Hobday et al., 2016 and Schlegel et al. 2021, but with only a 1-day persistence requirement. Positive values correspond to MHW categories, and negative values to MCS categories. Figure 12 (right) shows the coastal marine heat wave/cool spell flag tool that is provided as static images for the surface and bottom for the duration of the forecast window. Users can access the product via downloadable mp4's and png's, the interactive visualisation portal, incorporated into the raw, tier 1, tier 2 and tier 3 product netcdf files as well as in a timeseries file that provides the data for select 'virtual buoys' (Kleinsee, Hondeklipbaai, Doringbaai, Elandsbaai, Laaiplek, Paternoster, Saldanha", Yzerfontein, Bloubergstrand, Oudekraal, Cape Point, Simonstown, Strand, Hangklip, Kleinmond, Hermanus, Gansbaai).

Stratification

We provide a forecast proxy for stratification (see method above). Users can access the operational product via downloadable mp4's, the interactive visualisation portal, incorporated into the raw, tier 1, tier 2 and tier 3 product netcdf files as well as in a timeseries file that provides the data for select 'virtual buoys' (see above).

Sea Surface Temperature Fronts

Operational SST fronts are provided for the forecast window (-5 days to +3/ 5 days), calculated as the magnitude as the spatial temperature gradient (see equation below). Users can access the operational product via a downloadable mp4, the interactive visualisation portal, incorporated into the raw, tier 1, tier 2 and tier 3 product netcdf files as well as in a timeseries file that provides the data for select 'virtual buoys' (see above).

$$SST\ front = |\nabla T| = \sqrt{\left(\frac{\partial T}{\partial x}\right)^2 + \left(\frac{\partial T}{\partial y}\right)^2}$$

OpenDrift Tools

OpenDrift is an open source software package (<https://opendrift.github.io/>) for modelling the trajectories and fate of objects or substances drifting in the ocean. It has been integrated into the SOMISANA operational modelling workflow and is launched, on demand, for various offshore operations, including oil spill tracking and search and rescue operations. The SOMISANA OpenDrift functionality within the SOMISANA modelling workflow is also fully open-source on a git repo (<https://github.com/SAEON/somisana-opendrift>) that includes a fully documented wiki (<https://github.com/SAEON/somisana-opendrift/wiki>).

Harmful Algal Bloom Tracking

The monitoring of Harmful Algal Blooms (HABs) integrates ocean model forecasts with the OpenDrift functionality. The CSIR produces operational phytoplankton flags, based on chlorophyll concentrations from the Sentinel-3 Chl-a product, that identify various categories of blooms, with Flag 6 representing red tides. Our HAB tracking functionality seeds particles in OpenDrift where Flag 6 phytoplankton activity has been identified. These particles are subsequently advected by the

current field of the SA-West forecast model to produce 3- to 5-day forecasts of the movement of the HAB. Figure 13 provides an example case and validation of the application. Note that this tool does not include biogeochemical processes (i.e. growth and decay), it merely provides the probable advection pathway of an existing red tide. At present, this tracking capability is done on request.

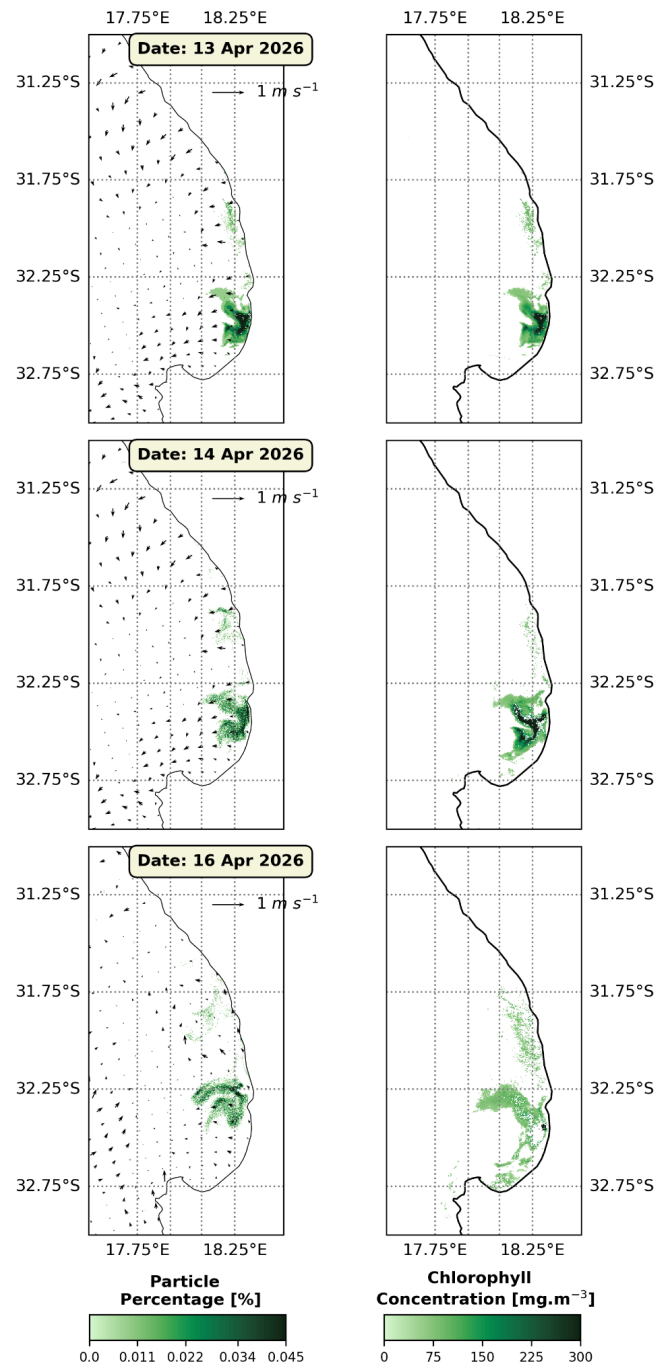


Figure 13: Left column: Percentage of particles (colour) with surface currents superimposed (black arrows; reference vector 1 m s⁻¹) showing the predicted advection of the Harmful Algal Bloom (HAB) from 13 April to 16 April 2026 using the SA-West Mercator-GFS model. Right column: Actual observed chlorophyll concentration (mg.m⁻³) of the HAB derived from satellite images.

Data Access

All data and outputs can be accessed through the SOMISANA catalogue (<https://catalog.somisana.ac.za/>) that points to the DFFE servers on which the datasets are stored:

Mnemosyne

All datasets can be navigated from here: <https://somisana.ocean.gov.za/sa-west/v1.0/>

Thredds

The hindcast datasets and products:

<https://somisana.ocean.gov.za/thredds/catalog/mnt/ocims-somisana/public-facing/sa-west/v1.0/hindcasts/GLORYS-ERA5/catalog.html>

The forecast datasets and products:

<https://somisana.ocean.gov.za/thredds/catalog/mnt/ocims-somisana/public-facing/sa-west/v1.0/forecasts/catalog.html>

Visualization Portal

An interactive visualization portal that allows users to explore the forecast outputs and value added products: https://data.ocean.gov.za/ocean_model_visualiser/

Table 4: Where to access various SA-West hindcast products

Product	Output type	Thredds	Mnemosyne	Visualization Portal	Available on Request
SA-West (ERA5) 1993-2025	• Monthly netcdf files (daily means) • Tier 1	✓	✓	✗	-
SA-West (WASA3) 2009-2013 Raw	• Monthly netcdf files (daily means) • Raw	✗	✗	✗	✓
SA-West (ERA5) Climatology	• Single netcdf file (daily means) • Tier 1	✓	✓	✗	-
SA-West (ERA5) Extremes (90th and 10th percentiles)	• Single netcdf file (daily) • Tier 1	✓	✓	✗	-

SA-West (ERA5) MHW/MCS 1993-2025	• Single netcdf file (daily) • Tier 1	✓	✓	✗	-
SA-West (ERA5) Stratification 1993-2025	• Single netcdf file (daily averages) • Tier 1	✓	✓	✗	-
SA-West (ERA5) Stratification Climatology	• Single netcdf file (daily means) • Tier 1	✓	✓	✗	-
SA-West (ERA5) Stratification Extremes (90th and 10th percentiles)	• Single netcdf file (daily) • Tier 1	✓	✓	✗	-

Table 5: Where to access various SA-West forecasts and value added products

Product	Output type	Thredds	Mnemosyne	Visualization Portal	Available on Request
SA-West forecast (all ensemble members)	• New netcdf files per forecast run (hourly for duration of run) • raw, Tier 1, Tier 2 and Tier 3	✓	✓	✓	-
SA-West forecast temperature anomalies (all ensemble members)	• Animation (mp4) • Surface, bottom and 100 m	✗	✓	✗	-
	• Variable in operational products netcdf file • raw, Tier 1, Tier 2 and Tier 3	✓	✓	✓	-
	• Variable in timeseries products netcdf file.	✓	✓	✗	-
SA-West MHW/MCS (all ensemble members)	• Animation • Surface and bottom	✗	✓	✗	-
	• Variable in operational products netcdf file • raw, Tier 1, Tier 2 and Tier 3	✓	✓	✓	-
	• Variable in timeseries products netcdf file.	✓	✓	✗	-

SA-West forecast stratification (all ensemble members)	• Animation (mp4)	✗	✓	✗	-
	• Variable in operational products netcdf file • raw, Tier 1, Tier 2 and Tier 3	✓	✓	✓	-
	• Variable in timeseries products netcdf file.	✓	✓	✗	-
SA-West forecast SST fronts (all ensemble members)	• Animation (mp4)	✗	✓	✗	-
	• Variable in operational products netcdf file • raw, Tier 1, Tier 2 and Tier 3	✓	✓	✓	-
	• Variable in timeseries products netcdf file.	✓	✓	✗	-
SA-West Drift tracking (all ensemble members)	• Animation (mp4) • Plots of particle density for the duration of the forecast window	✗	✗	✗	✓
	• Netcdf file • Drifter locations for the duration of the forecast run	✗	✗	✗	✓
SA-West HAB tracking (all ensemble members)	• Animation (mp4) • Plots of HAB advection for the duration of the forecast window	✗	✗	✗	✓
	• Netcdf file • Bloom location for the duration of the forecast run	✗	✗	✗	✓
SA-West Archived Forecasts	• Netcdf files	✗	✗	✗	✓

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